

PIC24 Encryption Machine

Midterm Presentation Summer 2026

Maric Joel Grundmann

Lukas Oliver Jung

Carlton Mukanda



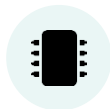
Progress Overview

A standalone AES-256-CBC encryption appliance has been implemented on the PIC24FJ256GB106 microcontroller. The firmware is fully modular, built with MPLAB Code Configurator and the XC16 compiler and reuses Microchip's MLA libraries for USB Host and FAT file access.



10

Firmware modules
implemented



256 KB

Flash on the
PIC24FJ256GB106



AES-256

CBC cipher engine
written in software



.hex

Build artifact
generated by MPLAB X



Hardware Architecture

Microcontroller

Device	PIC24FJ256GB106
Clock	12 MHz HS + PLL → 96 MHz
Flash / RAM	256 KB / 16 KB
Package	TQFP-64

Peripheral Connections



OLED – SH1101A

PMP (Parallel Master Port), 128×64 px



Touch pads (5×)

CTMU + ADC, AN8–AN12



RGB LED

PWM via OC1 / OC2 / OC3



USB Host

U1USB D+/D– (keyboard + mass storage)

Features delivered so far: OLED user interface, capacitive-touch and USB-keyboard input, read/write on USB mass storage (FAT16/32), AES-256-CBC encryption with a password-derived key and an RGB LED for status / error indication.



Prototype Preview – User Flow

Working interaction flow:

1



Power Up

Insert USB drive +
keyboard; display shows
“Ready”

2



Main Menu

Touch UP/DOWN to
choose Encrypt or Decrypt

3



Password Entry

Type password on USB
keyboard, confirm with
ENTER

4



File Selection

Scroll files (or .ENC files)
with touch pads

5



Processing

AES-256-CBC runs;
“Done” or “Error!” is
shown



Technical Development – Software Modules



main.c

Entry point; 6 CPU-trap handlers → RGB LED codes



menu.c

UI state machine, screens, touch-debounced input



cipher.c

AES-256-CBC engine + PKCS#7 padding



storage.c

FAT file system access via Microchip FILEIO



keyboard.c

USB HID keyboard driver with ring buffer



SH1101A.c

OLED display driver (128×64, PMP interface)



TouchSense.c

CTMU capacitive touch driver, 5 pads



RGBLeds.c

PWM-based RGB LED status control



usb_app.c

USB host application layer / event handling



usb_config.c

USB stack configuration (MSD + HID, TPL)



Cipher Core & Error Diagnostics

AES-256-CBC Core (cipher.c)

- CIPHER_Encrypt – AES-256-CBC over a buffer
- CIPHER_PKCS7_Pad / Unpad – padding handling
- KeyExpansion → 240 bytes of round keys
- SubBytes, ShiftRows, MixColumns, AddRoundKey
- S-box / Rcon tables stored in program flash
- Key material zeroed from the stack after use

Trap Handling → RGB LED (main.c)

Oscillator Fail	Yellow
Address Error	Red
Stack Error	Blue
Math Error	Green
DMA Error	Purple
Hard Trap	White

Six CPU fault traps each set a distinct LED color, then halt — giving an immediate, code-free diagnosis path.



Challenges & Next Steps



Current Blockers

- Encryption is not implemented yet
- Need to reset after every encryption
- Runs on specific keyboards



Planned Next Steps

- Implement existing encryption to the code
- Run an end-to-end encrypt/decrypt test on real hardware with the USB Host + FAT layer and a USB keyboard.



Build Status & Summary



Builds successfully

MPLAB X + XC16 v2.10 → .hex /
.elf in dist/



10 modules in place

Covering UI, crypto, storage, USB,
display, touch, LED



AES-256-CBC engine

Complete software
implementation with flash-
resident tables



GPLv3 + Apache 2.0

Custom code under GPLv3;
Microchip MLA libs under Apache
2.0

Thank you